

MS Module 5 Theory

Input models provide the driving force for a simulation model.

Input data are:

- In the queuing system, the distribution of time between arrivals and service times.
- For Inventory systems, the distributions are demand and lead time.
- For a reliability system, the distribution of time to failure of a component.

In real world simulation applications, however, determining appropriate distributions for input data is a major task from the standpoint of time and resource requirements.

Goodness-of-Fit Tests

Provides helpful (quantitative) guidance for evaluating the suitability of a potential input model

- Used in large samples size data
- Use tables to determine accept or reject

Conduct hypothesis testing on input data distribution using:

- Kolmogorov-Smirnov test
- Chi-square test

No single correct distribution in a real application exists.

- If very little data are available, it is unlikely to reject any candidate distributions
- If a lot of data are available, it is likely to reject all candidate distributions

One procedure for testing the hypothesis that a random sample of size n of the random variable X follows a specific distributional form is the chi-square goodness-of-fit test. This test formalizes the intuitive idea of comparing the histogram of the data to the shape of the candidate density or mass function. The test is valid for large sample sizes and for both discrete and continuous distributional assumptions, when parameters are estimated by maximum likelihood. The test procedure begins by arranging the n observations into a set of k class intervals or *cells*. The test statistic is given by

$$\chi_0^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \quad (9.17)$$

where O_i is the observed frequency in the i th class interval and E_i is the expected frequency in that class interval. The expected frequency for each class interval is computed as $E_i = np_i$, where p_i is the theoretical, hypothesized probability associated with the i th class interval.

It can be shown that χ_0^2 approximately follows the chi-square distribution with $k - s - 1$ degrees of freedom, where s represents the number of parameters of the hypothesized distribution estimated by the sample statistics. The hypotheses are the following:

H_0 : The random variable, X , conforms to the distributional assumption with the parameter(s) given by the parameter estimate(s).

H_1 : The random variable X does not conform.

If the distribution being tested is discrete, each value of the random variable should be a class interval, unless it is necessary to combine adjacent class intervals to meet the minimum-expected cell-frequency requirement. For the discrete case, if combining adjacent cells is not required,

$$p_i = p(x_i) = P(X = x_i)$$

Otherwise, p_i is found by summing the probabilities of appropriate adjacent cells.

Sample Size n	Number of Class Intervals k
20	Do not use the chi-square test
50	5 to 10
100	10 to 20
>100	$\sqrt{n}$ to $n/5$

If the distribution being tested is continuous, the class intervals are given by $[a_{i-1}, a_i)$, where a_{i-1} and a_i are the endpoints of the i th class interval. For the continuous case with assumed pdf $f(x)$, or assumed cdf $F(x)$, p_i can be computed as

$$p_i = \int_{a_{i-1}}^{a_i} f(x) dx = F(a_i) - F(a_{i-1})$$

For the discrete case, the number of class intervals is determined by the number of cells resulting after combining adjacent cells as necessary. However, for the continuous case, the number of class intervals must be specified. Although there are no general rules to be followed, the recommendations in Table 9.5 are made to aid in determining the number of class intervals for continuous data.

■ **Example Vehicle Arrival Example (Poisson Assumption)**

H_0 : the random variable is Poisson distributed
 H_1 : the random variable is not Poisson distributed

$$p(x) = \begin{cases} \frac{e^{-\alpha} \alpha^x}{x!}, & x = 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

$\chi^2_{0.05, 7-1-1} = 11.1 < 27.68$

For $\alpha = 3.64$, the probabilities associated with various values of x :

$P(0) = 0.028$	$P(4) = 0.192$	$P(8) = 0.020$
$P(1) = 0.096$	$P(5) = 0.140$	$P(9) = 0.008$
$P(2) = 0.174$	$P(6) = 0.085$	$P(10) = 0.003$
$P(3) = 0.211$	$P(7) = 0.044$	$P(11) = 0.001$

$E(x) = np$ →

2.6	19.2	2.0
9.6	14.0	0.8
17.4	8.5	0.3
21.1	4.4	0.1

Table 9.6 Chi-Square Goodness-of-Fit Test for Example 9.17

x_i	Observed Frequency O_i	Expected Frequency E_i	$\frac{(O_i - E_i)^2}{E_i}$
0	12	2.6	7.87
1	10	9.6	
2	19	17.4	
3	17	21.1	
4	10	19.2	
5	8	14.0	2.57
6	7	8.5	0.26
7	5	4.4	11.62
8	5	2.0	
9	3	0.8	
10	3	0.3	
≥ 11	1	0.1	
	<u>100</u>	<u>100.0</u>	<u>27.68</u>

From this information, Table 9.6 is constructed. The value of E_1 is given by $np_0 = 100(0.026) = 2.6$. In a similar manner, the remaining E_i values are computed. Since $E_1 = 2.6 < 5$, E_1 and E_2 are combined. In that case, O_1 and O_2 are also combined, and k is reduced by one. The last five class intervals are also combined, for the same reason, and k is further reduced by four.

The calculated χ_0^2 is 27.68. The degrees of freedom for the tabulated value of χ^2 is $k - s - 1 = 7 - 1 - 1 = 5$. Here, $s = 1$, since one parameter, $\hat{\alpha}$, was estimated from the data. At the 0.05 level of significance, the critical value $\chi_{0.05,5}^2$ is 11.1. Thus, H_0 would be rejected at level of significance 0.05. The analyst, therefore, might want to search for a better-fitting model or use the empirical distribution of the data.

Numericals:

■ Example 9.14 (Exponential Assumption)

H_0 : the random variable is Exponential distributed

H_1 : the random variable is not Exponential distributed

$$F(a_i) = 1 - e^{-\lambda a_i}$$

$$\hat{\lambda} = 1/\bar{X} = 0.084$$

Let $k = 8$, then each interval will have probability $p = 0.125$

$$ip = 1 - e^{-\lambda a_i} \Rightarrow a_i = -\frac{1}{\lambda} \ln(1 - ip)$$

$$a_1 = -\frac{1}{0.084} \ln(1 - 0.125) = 1.590$$

$$a_1 = 1.590, a_2 = 3.425, a_3 = 5.595, a_4 = 8.252, a_5 = 11.677, \\ a_6 = 16.503, a_7 = 24.755$$

Table 9.7 Chi-Square Goodness-of-Fit Test for Example 9.18

Class Interval	Observed Frequency O_i	Expected Frequency E_i	$\frac{(O_i - E_i)^2}{E_i}$
[0, 1.590)	19	6.25	26.01
[1.590, 3.425)	10	6.25	2.25
[3.425, 5.595)	3	6.25	0.81
[5.595, 8.252)	6	6.25	0.01
[8.252, 11.677)	1	6.25	4.41
[11.677, 16.503)	1	6.25	4.41
[16.503, 24.755)	4	6.25	0.81
[24.755, ∞)	6	6.25	0.01
	<u>50</u>	<u>50</u>	<u>39.6</u>

$$\chi_0^2 = 39.6 > \chi_{0.05, 8-1-1}^2 = 12.6$$

It is significant to reject H_0 at the 0.05 level of significance.

Kolmogorov Smirnov Test:

Intuition: formalize the idea behind examining a $q-q$ plot

A more powerful test, particularly useful when:

- Sample sizes are small,
- No parameters have been estimated from the data.

When parameter estimates have been made:

- Critical values are biased, too large.
- More conservative, i.e., smaller Type I error than specified.

p-Values and “Best Fits”

- p-value for the test statistics
 - The significance level at which one would just reject H_0 for the given test statistic value.
 - A measure of fit, the larger the better
 - Large p-value: good fit
 - Small p-value: poor fit
- Vehicle Arrival Example (cont.):
 - H_0 : data is Poisson
 - Test statistics: $\chi^2=27.68$, with 5 degrees of freedom
 - p-value = 0.00004, meaning we would reject H_0 with 0.00004 significance level, hence Poisson is a poor fit.

Many software use p-value as the ranking measure to automatically determine the “best fit”.

Things to be cautious about:

- Software may not know about the physical basis of the data, distribution families it suggests may be inappropriate.
- Close conformance to the data does not always lead to the most appropriate input model.
- p-value does not say much about where the lack of fit occurs.

Selecting Model without Data:

If data is not available, some possible sources to obtain information about the process are:

- Engineering data: often product or process has performance ratings provided by the manufacturer or company rules specify time or production standards.
(For example, a laser printer can produce 4 pages/minute)
- Expert option: Talk to people who are experienced with the process or similar processes. Often, they can provide optimistic, pessimistic, and most-likely times, and they may know the variability as well.
- Physical or conventional limitations: physical limits on performance, limits or bounds that narrow the range of the input process.
(For example, computer data entry cannot be faster than a person can type)

The nature of the process: Select the family of distribution

Note: When data are not available, the uniform, triangular, and beta distributions are often used

as input models.

■ **Example: Production planning simulation.**

□ Input of sales volume of various products is required, salesperson of product XYZ says that:

- No fewer than 1,000 units and no more than 5,000 units will be sold.
- Given her experience, she believes there is a 90% chance of selling more than 2,000 units, a 25% chance of selling more than 2,500 units, and only a 1% chance of selling more than 4,500 units.

□ Translating these information into a cumulative probability of being less than or equal to those goals for simulation input:

i	Interval (Sales)	Cumulative Frequency, c_i
1	$1000 \leq x \leq 2000$	0.10
2	$2000 < x \leq 3000$	0.75
3	$3000 < x \leq 4000$	0.99
4	$4000 < x \leq 5000$	1.00

9.7.1 Covariance and Correlation

Let X_1 and X_2 be two random variables, and let $\mu_i = E(X_i)$ and $\sigma_i^2 = V(X_i)$ be the mean and variance of X_i , respectively. The *covariance* and *correlation* are measures of the linear dependence between X_1 and X_2 . In other words, the covariance and correlation indicate how well the relationship between X_1 and X_2 is described by the model

$$(X_1 - \mu_1) = \beta(X_2 - \mu_2) + \varepsilon$$

where ε is a random variable with mean 0 that is independent of X_2 . If, in fact, $(X_1 - \mu_1) = \beta(X_2 - \mu_2)$, then this model is perfect. On the other hand, if X_1 and X_2 are statistically independent, then $\beta = 0$ and the model is of no value. In general, a positive value of β indicates that X_1 and X_2 tend to be above or below their means together; a negative value of β indicates that they tend to be on opposite sides of their means.

The covariance between X_1 and X_2 is defined to be

$$\text{cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E(X_1 X_2) - \mu_1 \mu_2 \quad (9.23)$$

The value $\text{cov}(X_1, X_2) = 0$ implies $\beta = 0$ in our model of dependence, and $\text{cov}(X_1, X_2) < 0$ (> 0) implies $\beta < 0$ (> 0).

The covariance can take any value between $-\infty$ and ∞ . The correlation standardizes the covariance to be between -1 and 1 :

$$\rho = \text{corr}(X_1, X_2) = \frac{\text{cov}(X_1, X_2)}{\sigma_1 \sigma_2} \quad (9.24)$$

- A time series is a sequence of random variables $X_1, X_2, X_3, \dots$, are identically distributed (same mean and variance) but dependent.
 - $\text{cov}(X_t, X_{t+h})$ is the lag- h autocovariance
 - $\text{corr}(X_t, X_{t+h})$ is the lag- h autocorrelation
 - If the autocovariance value depends only on h and not on t , the time series is covariance stationary

Multivariate:

- For example Lead time and annual demand for an inventory model increase in demands results in Lead time increase hence variables are dependent

- If X_1 and X_2 are normally distributed, dependence between them can be modeled by the bivariate normal distribution with $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ and correlation ρ
 - To Estimate $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$, see "Parameter Estimation" as seen earlier
 - To Estimate ρ , suppose we have n independent and identically distributed pairs $(X_{11}, X_{21}), (X_{12}, X_{22}), \dots, (X_{1n}, X_{2n})$, then:

$$\begin{aligned}\hat{\text{cov}}(X_1, X_2) &= \frac{1}{n-1} \sum_{j=1}^n (X_{1j} - \hat{X}_1)(X_{2j} - \hat{X}_2) \\ &= \frac{1}{n-1} \left(\sum_{j=1}^n X_{1j} X_{2j} - n \hat{X}_1 \hat{X}_2 \right)\end{aligned}$$

$$\hat{\rho} = \frac{\hat{\text{cov}}(X_1, X_2)}{\hat{\sigma}_1 \hat{\sigma}_2}$$

Sample deviation

Time series:

- for example time between intervals of Orders to buy and sell stocks
- time between arrivals are dependent

- If $X_1, X_2, X_3, \dots$ is a sequence of identically distributed, but dependent and covariance-stationary random variables, then we can represent the process as follows:

- Autoregressive order-1 model, AR(1)
- Exponential autoregressive order-1 model, EAR(1)

- Both have the characteristics that:

$$\rho_h = \text{corr}(X_t, X_{t+h}) = \rho^h, \quad \text{for } h = 1, 2, \dots$$

- Lag- h autocorrelation decreases geometrically as the lag increases, hence, observations far apart in time are nearly independent

AR(1) MODEL

Consider the time-series model

$$X_t = \mu + \phi(X_{t-1} - \mu) + \varepsilon_t \quad (9.27)$$

for $t = 2, 3, \dots$, where $\varepsilon_2, \varepsilon_3, \dots$ are independent and identically (normally) distributed with mean 0 and variance σ_ε^2 , and $-1 < \phi < 1$. If the initial value X_1 is chosen appropriately (see below), then $X_1, X_2, \dots$ are all normally distributed with mean μ , variance $\sigma_\varepsilon^2/(1 - \phi^2)$, and

$$\rho_h = \phi^h$$

for $h = 1, 2, \dots$. This time-series model is called the *autoregressive order-1 model*, or AR(1) for short.

Estimation of the parameter ϕ can be obtained from the fact that

$$\phi = \rho^1 = \text{corr}(X_t, X_{t+1})$$

the lag-1 autocorrelation. Therefore, to estimate ϕ , we first estimate the lag-1 autocovariance by

$$\begin{aligned} \widehat{\text{cov}}(X_t, X_{t+1}) &= \frac{1}{n-1} \sum_{t=1}^{n-1} (X_t - \bar{X})(X_{t+1} - \bar{X}) \\ &= \frac{1}{n-1} \left(\sum_{t=1}^{n-1} X_t X_{t+1} - (n-1)\bar{X}^2 \right) \end{aligned} \quad (9.28)$$

and the variance $\sigma^2 = \text{var}(X)$ by the usual estimator $\hat{\sigma}^2$. Then

$$\hat{\phi} = \frac{\widehat{\text{cov}}(X_t, X_{t+1})}{\hat{\sigma}^2}$$

Finally, estimate μ and σ_ε^2 by $\hat{\mu} = \bar{X}$ and $\hat{\sigma}_\varepsilon^2 = \hat{\sigma}^2(1 - \hat{\phi}^2)$, respectively.

The following algorithm generates a stationary AR(1) time series, given values of the parameters ϕ , μ , and σ_ε^2 :

- Step 1.** Generate X_1 from the normal distribution with mean μ and variance $\sigma_\varepsilon^2/(1 - \phi^2)$. Set $t = 2$.
- Step 2.** Generate ε_t from the normal distribution with mean 0 and variance σ_ε^2 .
- Step 3.** Set $X_t = \mu + \phi(X_{t-1} - \mu) + \varepsilon_t$.
- Step 4.** Set $t = t + 1$ and go to Step 2.

EAR(1) MODEL

Consider the time-series model

$$X_t = \begin{cases} \phi X_{t-1}, & \text{with probability } \phi \\ \phi X_{t-1} + \varepsilon_t, & \text{with probability } 1 - \phi \end{cases} \quad (9.29)$$

for $t = 2, 3, \dots$, where $\varepsilon_2, \varepsilon_3, \dots$ are independent and identically (exponentially) distributed with mean $1/\lambda$ and $0 \leq \phi < 1$. If the initial value X_1 is chosen appropriately (see below), then $X_1, X_2, \dots$ are all exponentially distributed with mean $1/\lambda$ and

$$\rho_h = \phi^h$$

for $h = 1, 2, \dots$. This time-series model is called the *exponential autoregressive order-1 model*, or EAR(1) for short. Only autocorrelations greater than 0 can be represented by this model. Estimation of the parameters proceeds as for the AR(1) by setting $\hat{\phi} = \hat{\rho}$, the estimated lag-1 autocorrelation, and setting $\hat{\lambda} = 1/\bar{X}$.

The following algorithm generates a stationary EAR(1) time series, given values of the parameters ϕ and λ :

Step 1. Generate X_1 from the exponential distribution with mean $1/\lambda$. Set $t = 2$.

Step 2. Generate U from the uniform distribution on $[0, 1]$. If $U \leq \phi$, then set

$$X_t = \phi X_{t-1}$$

Otherwise, generate ε_t from the exponential distribution with mean $1/\lambda$ and set

$$X_t = \phi X_{t-1} + \varepsilon_t$$

Step 3. Set $t = t + 1$ and go to Step 2.

Example 9.24: Example 9.22 (cont.)

The stock broker would typically have a large sample of data, but, for the sake of illustration, suppose that the following twenty time gaps between customer buy and sell orders had been recorded (in seconds): 1.95, 1.75, 1.58, 1.42, 1.28, 1.15, 1.04, 0.93, 0.84, 0.75, 0.68, 0.61, 11.98, 10.79, 9.71, 14.02, 12.62, 11.36, 10.22, 9.20. Standard calculations give $\bar{X} = 5.2$ and $\hat{\sigma}^2 = 26.7$. To estimate the lag-1 autocorrelation, we need

$$\sum_{j=1}^{19} X_j X_{j+1} = 924.1$$

Thus, $\widehat{\text{cov}} = [924.1 - (20 - 1)(5.2)^2]/(20 - 1) = 21.6$, and

$$\hat{\rho} = \frac{21.6}{26.7} = 0.8$$

Therefore, we could model the interarrival times as an EAR(1) process with $\hat{\lambda} = 1/5.2 = 0.192$ and $\hat{\phi} = 0.8$, provided that an exponential distribution is a good model for the individual gaps.

Purpose & Overview

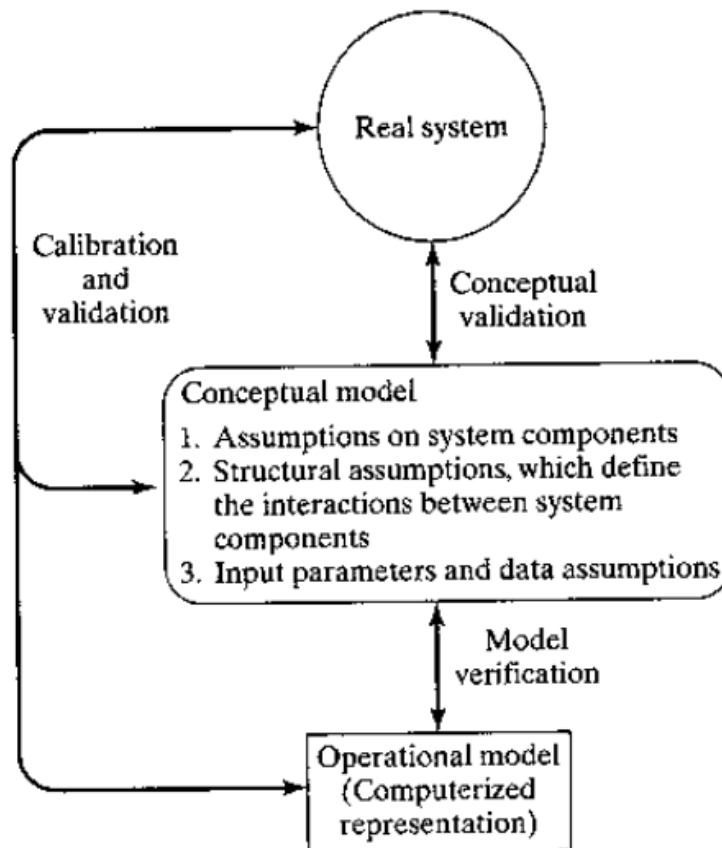
The goal of the validation process is:

- To produce a model that represents true behavior closely enough for decision-making purposes
- To increase the model's credibility to an acceptable level Validation is an integral part of model development

Verification-building the model correctly (correctly implemented with good input and structure)

Validation-building the correct model (an accurate representation of the real system)

Most methods are informal subjective comparisons while a few are formal statistical procedures



Model building

- First step: observe the real time system and interactions among its various components and collecting data on its behavior
- Second step construction of conceptual model which include collection of assumptions on components and structure of the model, hypothesis on values of model input parameters
- Third step: translational of operational model to computer recognizable form-computerized model.

Verification

Purpose: ensure the conceptual model is reflected accurately in the computerized representation.
Many common-sense suggestions, for example: Have someone else check the model.

- Make a flow diagram that includes each logically possible action a system can take when an event occurs.
- Closely examine the model output for reasonableness under a variety of input parameter settings. (Often overlooked!)
- Print the input parameters at the end of the simulation, make sure they have not been changed inadvertently.
- The interactive run controller (IRC) or debugger is an essential component for simulation model building graphical interfaces for verification and validation

Examination of Model Output for Reasonableness

Example: A model of a complex network of queues consisting of many service centers.

Response time is the primary interest, however, it is important to collect and print out many statistics in addition to response time.

Two statistics that give a quick indication of model reasonableness are current contents and total counts, for example:

- If the current content grows in a more or less linear fashion as the simulation run time increases, it is likely that a queue is unstable
- If the total count for some subsystem is zero, indicates no items entered that subsystem, a highly suspect occurrence
- If the total and current count are equal to one, can indicate that an entity has captured a resource but never freed that resource.
- If the current contents in some part of the system is high, a large number of transactions are delayed

Compute certain long-run measures of performance, e.g. compute the long-run server utilization and compare to simulation results

Other Important Tools

Documentation

- A means to aid the verification process
- Model builder must include:
 - Comments for the computerized model
 - Definition of variables
- Description of each section of computerized model: This helps in clarifying the logic of a model and verifying its completeness

Use of a trace

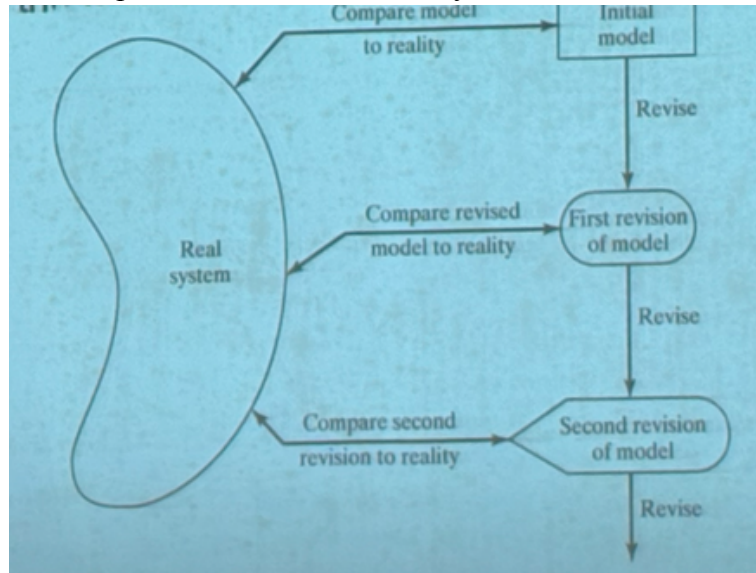
- A detailed printout of the state of the simulation model over time.
- To get values of selected variables as simulation clock advances
- Some:

- Trace set for specific location
- Trace set for particular activity
- Trace set for the occurrence of a particular condition

Calibration and Validation

- **Validation:** the overall process of comparing the model and its behavior to the real system.
- **Calibration:** the iterative process of comparing the model to the real system and making adjustments.

It adjusts the model and its parameters to best fit the system behavior



Three-step approach (Naylor and Finger)

- Build a model that has high face validity.
- Validate model assumptions.
- Compare the model input-output transformations with the real system's data.

High Face Validity

- Construct a model that appears reasonable on its face (ensure high degree of realism) to model users and experts i.e. Ensure a high degree of realism into the model through reasonable assumptions regarding system structure and reliable data
- Potential users should be involved in: model construction (from its conceptualization to its implementation). Evaluation of model output for reasonableness
- Identification of model deficiencies
- User involvement also increases the model's perceived validity or credibility

Validate Input-Output Transformation

Goal: Validate the model's ability to predict future behavior

The only objective test of the model:

- The structure of the model should be accurate enough to make good predictions for the range of input data sets of interest.

One possible approach: use historical data that have been reserved for validation purposes only.

- Criteria: use the main responses of interest.
- Validate in various situations
- Planning stage
- Subsystem

Validation of non existing newly proposed system: two models under similar input conditions

If the model is a modification of existing system

- Minor changes of single numerical parameters
- Arrival time, no of servers
- Minor changes in form of statistical distribution
- Distribution of service time
- Major changes in structure
- Queue discipline

Major change in different design of new system Non computer based to computerised

Example 10.2: The Fifth National Bank of Jaspar

The Fifth National Bank of Jaspar, as shown in Figure 10.4, is planning to expand its drive-in service at the corner of Main Street. Currently, there is one drive-in window serviced by one teller. Only one or two transactions are allowed at the drive-in window, so it was assumed that each service time was a random sample from some underlying population. Service times $\{S_i, i = 1, 2, \dots, 90\}$ and interarrival times $\{A_i, i = 1, 2, \dots, 90\}$ were collected for the 90 customers who arrived between 11:00 A.M. and 1:00 P.M. on a Friday. This time slot was selected for data collection after consultation with management and the teller because it was felt to be representative of a typical rush hour.

Data analysis (as outlined in Chapter 9) led to the conclusion that arrivals could be modeled as a Poisson process at a rate of 45 customers per hour and that service times were approximately normally distributed, with mean 1.1 minutes and standard deviation 0.2 minute. Thus, the model has two input variables:

1. interarrival times, exponentially distributed (i.e., a Poisson arrival process) at rate $\lambda = 45$ per hour;
2. service times, assumed to be $N(1.1, (0.2)^2)$.

$$\begin{array}{l} H_0 : E(Y_2) = 4.3 \text{ minutes} \\ \text{versus} \\ H_1 : E(Y_2) \neq 4.3 \text{ minutes} \end{array} \quad (10.1)$$

is conducted. If H_0 is not rejected, then, on the basis of this test, there is no reason to consider the model invalid. If H_0 is rejected, the current version of the model is rejected, and the modeler is forced to seek ways to improve the model, as illustrated by Figure 10.3. As formulated here, the appropriate statistical test is the t test, which is conducted in the following manner:

Choose a level of significance α and a sample size n . For the bank model, choose

$$\alpha = 0.05, \quad n = 6$$

Compute the sample mean $\bar{Y}_2$ and the sample standard deviation S over the n replications, by using Equations (9.1) and (9.2):

and

$$S = \left[\frac{\sum_{i=1}^n (Y_{2i} - \bar{Y}_2)^2}{n-1} \right]^{1/2} = 0.82 \text{ minute}$$

where Y_{2i} , $i = 1, \dots, 6$, are as shown in Table 10.2.

Get the critical value of t from Table A.5. For a two-sided test, such as that in equation (10.1), use $t_{\alpha/2, n-1}$; for a one-sided test, use $t_{\alpha, n-1}$ or $-t_{\alpha, n-1}$, as appropriate ($n-1$ being the degrees of freedom). From Table A.5, $t_{0.025, 5} = 2.571$ for a two-sided test.

Table 10.2 Results of Six Replications of the Bank Model

Replication	Y_4 (arrivals/hour)	Y_5 (minutes)	$Y_2 = \text{Average Delay}$ (minutes)
1	51	1.07	2.79
2	40	1.12	1.12
3	45.5	1.06	2.24
4	50.5	1.10	3.45
5	53	1.09	3.13
6	49	1.07	2.38
Sample mean			2.51
Standard deviation			0.82

Compute the test statistic

$$t_0 = \frac{\bar{Y}_2 - \mu_0}{S/\sqrt{n}} \quad (10.2)$$

where μ_0 is the specified value in the null hypothesis H_0 . Here $\mu_0 = 4.3$ minutes, so that

$$t_0 = \frac{2.51 - 4.3}{0.82/\sqrt{6}} = -5.34$$

For the two-sided test, if $|t_0| > t_{\alpha/2, n-1}$, reject H_0 . Otherwise, do not reject H_0 . [For the one-sided test with $H_1 : E(Y_2) > \mu_0$, reject H_0 if $t > t_{\alpha, n-1}$; with $H_1 : E(Y_2) < \mu_0$, reject H_0 if $t < -t_{\alpha, n-1}$.]

Since $|t| = 5.34 > t_{0.025, 5} = 2.571$, reject H_0 , and conclude that the model is inadequate in its prediction of average customer delay.

Recall that, in the testing of hypotheses, rejection of the null hypothesis H_0 is a strong conclusion, because

$$P(H_0 \text{ rejected} | H_0 \text{ is true}) = \alpha \quad (10.3)$$

Power of test

- The power of test is the probability of detecting a departure from $H_0 : \mu = \mu_0$ when in fact such a departure exists.
- The power of test is the probability of detecting an invalid model

Table 10.4 Types of Error in Model Validation

<i>Statistical Terminology</i>	<i>Modeling Terminology</i>	<i>Associated Risk</i>
Type I: rejecting H_0 when H_0 is true	Rejecting a valid model	α
Type II: failure to reject H_0 when H_1 is true	Failure to reject an invalid model	β

Using Historical Output Data

An alternative to generating input data: Use the actual historical record.

- Drive the simulation model with the historical record and then compare model output to system data.
- In the bank example, use the recorded interarrival and service times for the customers $\{A_n, S_n, n = 1, 2, \dots\}$.
- Procedure and validation process: similar to the approach used for system generated input data.

Using a Turing Test

- Used in addition to statistical test, or when no statistical test is readily applicable.
- Utilize persons' knowledge about the system.

For example:

- Present 10 system performance reports to a manager of the system. Five of them are from the real system and the rest are "fake" reports based on simulation output data.
- If the person identifies a substantial number of the fake reports, interview the person to get information for model improvement. If the person cannot distinguish between fake and real reports with consistency, conclude that the test gives no evidence of model inadequacy.

Outline

- Distinguish the two types of simulation: transient vs. steady state.
- Illustrate the inherent variability in a stochastic discrete- event simulation.
- Cover the statistical estimation of performance measures.

Output analysis is the examination of the data generated by simulation

Objective:

- Estimate system performance of a single system via simulation
- Compare performance of two or more alternative system design

The need for statistical output analysis is based on the observation that the output data from the simulation exhibits random variability

If θ is the system performance, the precision of the estimator $\hat{\theta}$ can be measured by:

- The standard error of $\hat{\theta}$.
- The width of a confidence interval (CI) for θ .

Purpose of statistical analysis:

- To estimate the standard error or CI .
- To figure out the number of observations required to achieve desired error/CI.

Potential issues to overcome:

- Autocorrelation of output variables so statistical tests are not directly applicable , e.g. inventory cost for subsequent weeks lack statistical independence.
- Initial conditions , may influence output data.
- e.g. inventory on hand and no. of backorders at time 0 would most likely influence the performance of week 1.

Type of Simulations

- Terminating verses non-terminating simulations
- Terminating simulation:
 - Runs for some duration of time T_E , where E is a specified event that stops the simulation.
 - Starts at time 0 under well-specified initial conditions.
 - Ends at the stopping time T_E .
 - Bank example: Opens at 8:30 am (time 0) with no customers present and 8 of the 11 teller working (initial conditions), and closes at 4:30 pm (Time $T_E = 480$ minutes).
 - The simulation analyst chooses to consider it a terminating system because the object of interest is one day's operation.
 - Transient simulation or finite horizon

Type of Simulations

Non-terminating simulation:

- Runs continuously, or at least over a very long period of time.

Examples: assembly lines that shut down infrequently, telephone systems, hospital emergency rooms.

- Initial conditions defined by the analyst.
- Runs for some analyst-specified period of time TE.
- Whether a simulation is considered to be terminating or non-terminating depends on both
- The objectives of the simulation study and the nature of the system.

Stochastic Nature of Output Data

- Model output consist of one or more random variables (r. v.) because the model is an input-output transformation and the input variables are r.v.'s.
- M/G/1 queueing example:
 - Poisson arrival rate = 0.1 per minute; service time $\sim N(\mu = 9.5, \sigma = 1.75)$.
 - System performance: long-run mean queue length, $L_Q(t)$.
 - Suppose we run a single simulation for a total of 5,000 minutes
 - Divide the time interval $[0, 5000)$ into 5 equal subintervals of 1000 minutes.
 - Average number of customers in queue from time $(j-1)1000$ to $j(1000)$ is Y_j .

Two issues addressed by statistical analysis

- Output data from various runs are independent and identically distributed
- The effects of correlation and initial conditions of the estimation of the long run mean measures of performance.

Measures of performance

- Consider the estimation of a performance parameter, θ (or ϕ), of a simulated system.
 - Discrete time data: $[Y_1, Y_2, \dots, Y_n]$, with ordinary mean: θ
 - Continuous-time data: $\{Y(t), 0 \leq t \leq T_E\}$ with time-weighted mean: ϕ
- Point estimation for discrete time data.
 - The point estimator:
$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n Y_i$$
 - Is unbiased if its expected value is θ , that is if: $E(\hat{\theta}) = \theta$
 - Is biased if: $E(\hat{\theta}) \neq \theta$
 - The estimator bias is $E(\hat{\theta}) - \theta$

Desired

Quantile and Percentile Estimator:

- Performance measure that does not fit: quantile or percentile:
 - Let W_Q represent delay in the Queue
 - The 0.75 quantile of W_Q is the value θ such that
$$\Pr\{W \leq \theta\} = p \quad \text{where } p=0.75$$
 - As a percentage, θ is the 100th or 75th percentile of customer delay
 - Hence 75% of all the customers will experience a delay of θ minutes or less
- Median is widely used as performance measure

11.3.2 Confidence-Interval Estimation

To understand confidence intervals fully, it is important to understand the difference between a *measure of error* and a *measure of risk*. One way to make the difference clear is to contrast a *confidence interval* with a *prediction interval* (which is another useful output-analysis tool).

Both confidence intervals and prediction intervals are based on the premise that the data being produced by the simulation is represented well by a probability model. Suppose that model is the normal distribution with mean θ and variance σ^2 , both unknown. To make the example concrete, let Y_i be the average cycle time for parts produced on the i th replication (representing a day of production) of the simulation. Therefore, θ is the mathematical expectation of Y_i , and σ represents the day-to-day variation of the average cycle time.

Suppose our goal is to estimate θ . If we are planning to be in business for a long time, producing parts day after day, then θ is a relevant parameter, because it is the long-run mean daily cycle time. Our average cycle time will vary from day to day, but over the long run the average of the averages will be close to θ .

The natural estimator for θ is the overall sample mean of n independent replications, $\bar{Y} = \sum_{i=1}^n Y_i / n$. But $\bar{Y}$ is not θ , it is an estimate, based on a sample, and it has error. A confidence interval is a measure of that error. Let

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

be the sample variance across the R replications. The usual confidence interval, which assumes the Y_i are normally distributed, is

$$\bar{Y} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

where $t_{\alpha/2, R-1}$ is the quantile of the t distribution with $R - 1$ degrees of freedom that cuts off $\alpha/2$ of the area of each tail. (See Table A.5.) We cannot know for certain exactly how far $\bar{Y}$ is from θ , but the confidence interval attempts to bound that error. Unfortunately, the confidence interval itself may be wrong. A confidence level, such as 95%, tells us how much we can trust the interval to actually bound the error between $\bar{Y}$ and θ . The more replications we make, the less error there is in $\bar{Y}$, and our confidence interval reflects that because $t_{\alpha/2, n-1}S/\sqrt{n}$ will tend to get smaller as n increases, converging to 0 as n goes to infinity.

The normal-theory prediction interval is

$$\bar{Y} \pm t_{\alpha/2, n-1}S\sqrt{1 + \frac{1}{n}}$$

The length of this interval will not go to 0 as n increases. In fact, in the limit it becomes

$$\theta \pm z_{\alpha/2}\sigma$$

- Median is 0.5 quantile or 50th percentile
 - Estimating quantiles: the inverse of the problem of estimating a proportion or probability.
 - Consider a histogram of the observed values Y :
 - Find $\hat{\theta}$ such that 100 p % of the histogram is to the left of (smaller than) $\hat{\theta}$.
- Let us consider delay from 150 customers.
Now the estimate of 75th percentile of delay
=0.75 of 150 =112.5 =113
Hence 113 observed values are less than or equal to
 θ and $\hat{\theta}$ is equal to 113th smallest value to given sample*

Statistical Background

[Terminating Simulations]

- Important to distinguish within-replication data from across-replication data.
- For example, simulation of a manufacturing system
 - Two performance measures of that system:
 - cycle time for parts and
 - work in process (WIP).
 - Let Y_{ij} be the cycle time for the j^{th} part produced in the i^{th} replication.
 - Across-replication data are formed by summarizing within-replication data
 - $\bar{Y}_i$ is the sample mean of the n_i cycle times from the i^{th} replication and S_i^2 is variance

■ Across Replication:

- For example: the daily cycle time averages (discrete time data)

- The average: $\bar{Y} = \frac{1}{R} \sum_{i=1}^R Y_i$

- The sample variance: $S^2 = \frac{1}{R-1} \sum_{i=1}^R (Y_i - \bar{Y})^2$

- The confidence-interval half-width: $H = t_{\alpha/2, R-1} \frac{S}{\sqrt{R}}$

■ Within replication:

- For example: the WIP (a continuous time data)

- The average: $\bar{Y}_i = \frac{1}{T_{Ei}} \int_0^{T_{Ei}} Y_i(t) dt$

- The sample variance: $S_i^2 = \frac{1}{T_{Ei}} \int_0^{T_{Ei}} (Y_i(t) - \bar{Y}_i)^2 dt$

- To understand confidence intervals fully, it is important to distinguish between measures of error, and measures of risk, e.g., confidence interval versus prediction interval.

- Suppose the model is the normal distribution with mean θ , variance σ^2 (both unknown).

- Let Y_i be the average cycle time for parts produced on the i^{th} replication of the simulation (its mathematical expectation is θ).
 - Average cycle time will vary from day to day, but over the long-run the average of the averages will be close to θ .
 - Sample variance across R replications:

$$S^2 = \frac{1}{R-1} \sum_{i=1}^R (Y_i - \bar{Y})^2$$

Call Center Example: estimate the agent's utilization ρ over the first 2 hours of the workday.

- Initial sample of size $R_0 = 4$ is taken and an initial estimate of the population variance is $S_0^2 = (0.072)^2 = 0.00518$.
- The error criterion is $\varepsilon = 0.04$ and confidence coefficient is $1-\alpha = 0.95$, hence, the final sample size must be at least:

$$\left(\frac{z_{0.025} S_0}{\varepsilon} \right)^2 = \frac{1.96^2 * 0.00518}{0.04^2} = 12.14$$

- For the final sample size:

R	13	14	15
$t_{0.025, R-1}$	2.18	2.16	2.14
$(t_{\alpha/2, R-1} S_0 / \varepsilon)^2$	15.39	15.1	14.83

- $R = 15$ is the smallest integer satisfying the error criterion, so $R - R_0 = 11$ additional replications are needed.
- After obtaining additional outputs, half-width should be checked.

Quantiles:

When the number of independent replications $Y_1, \dots, Y_R$ is large enough that $t_{\alpha/2, R-1} = z_{\alpha/2}$, the confidence interval for a probability p is often written as:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{R-1}}$$

The sample proportion

A quantile is the inverse of the probability to the probability estimation problem:

Find θ such that $Pr(Y \leq \theta) = p$ $\leftarrow p$ is given

Output Analysis for Steady state Simulation:

Consider a single run of a simulation model whose purpose is to estimate a steady-state, or long-run, characteristic of the system. Suppose that the single run produces observations $Y_1, Y_2, \dots$, which,

generally, are samples of an autocorrelated time series. The steady-state (or long-run) measure of performance, θ , is defined by

$$\theta = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n Y_i \quad (11.19)$$

with probability 1, where the value of θ is independent of the initial conditions. (The phrase “with probability 1” means that essentially all simulations of the model, using different random numbers, will produce a series $Y_i, i = 1, 2, \dots$ whose sample average converges to θ .) For example, if Y_i was the time customer i spent talking to an operator, then θ would be the long-run average time a customer spends talking to an operator; and, because θ is defined as a limit, it is independent of the call center’s conditions at time 0. Similarly, the steady-state performance for a continuous-time output measure $\{Y(t), t \geq 0\}$, such as the number of customers in the call center’s hold queue, is defined as

$$\phi = \lim_{T_E \rightarrow \infty} \frac{1}{T_E} \int_0^{T_E} Y(t) dt$$

with probability 1.

Initialization Bias

Methods to reduce the point-estimator bias caused by using artificial and unrealistic initial conditions:

- Intelligent initialization.
- Divide simulation into an initialization phase and data-collection phase.

Intelligent initialization

- Initialize the simulation in a state that is more representative of long-run conditions.
- If the system exists, collect data on it and use these data to specify more nearly typical initial conditions.
- If the system can be simplified enough to make it mathematically solvable, e.g. queueing models, solve the simplified model to find long-run expected or most likely conditions, use that to initialize the simulation.

■ **Divide each simulation into two phases:**

- An initialization phase, from time 0 to time T_0 .
- A data-collection phase, from T_0 to the stopping time $T_0 + T_E$.
- The choice of T_0 is important:
 - After T_0 , system should be more nearly representative of steady-state behavior.
- System has reached steady state: the probability distribution of the system state is close to the steady-state probability distribution (bias of response variable is negligible).

■ **Warming up of model/Initial data deletion**

■ **Cumulative average sample mean (after deleting d observations):**

$$\bar{Y}(n, d) = \frac{1}{n-d} \sum_{j=d+1}^n \bar{Y}_j$$

No widely accepted, objective and proven technique to guide how much data to delete to reduce initialization bias to a negligible level.

Plots can, at times, be misleading but they are still recommended.

- Ensemble averages reveal a smoother and more precise trend as the # of replications, R , increases.
- Ensemble averages can be smoothed further by plotting a moving average.
- Cumulative average becomes less variable as more data are averaged.
- The more correlation present, the longer it takes for $\bar{Y}$, to approach steady state.
- Different performance measures could approach steady state at different rates.

Error Estimation:

If $\{Y_1, \dots, Y_n\}$ are not statistically independent, then S^2/n is a biased estimator of the true variance. Almost always the case when $\{Y_1, \dots, Y_n\}$ is a sequence of output observations from within a single replication (autocorrelated sequence, time-series).

Suppose the point estimator is the sample mean

- $\bar{Y} = \sum Y/n$

Variance of $\bar{y}$ is almost impossible to estimate.

For systems with steady state, produce an output process that is approximately covariance stationary (after passing the transient phase).

- The covariance between two random variables in the time series depends only on the lag (the # of observations between them).

- If $\{Y_1, \dots, Y_n\}$ are not statistically independent, then S^2/n is a biased estimator of the true variance.

- Almost always the case when $\{Y_1, \dots, Y_n\}$ is a sequence of output observations from within a single replication (autocorrelated sequence, time-series).

- Suppose the point estimator $\bar{\theta}$ is the sample mean

$$\bar{Y} = \sum_{i=1}^n Y_i / n$$

- Variance of $\bar{Y}$ is almost impossible to estimate.
- For system with steady state, produce an output process that is approximately **covariance stationary** (after passing the transient phase).
 - The covariance between two random variables in the time series depends only on the lag (the # of observations between them).

- For a covariance stationary time series, $\{Y_1, \dots, Y_n\}$:

- Lag-k autocovariance is: $\delta_k = \text{cov}(Y_1, Y_{1+k}) = \text{cov}(Y_i, Y_{i+k})$

- Lag-k autocorrelation is: $\rho_k = \frac{\gamma_k}{\sigma^2}$

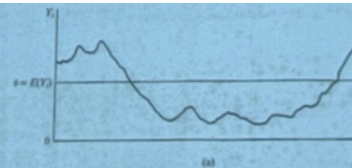
- If a time series is covariance stationary, then the variance of $\bar{Y}$ is:

$$V(Y) = \frac{\sigma^2}{n} \left[1 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n} \right) \rho_k \right]$$

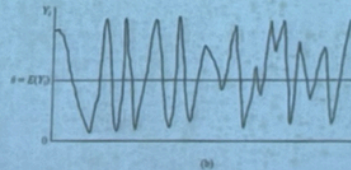
- The expected value of the variance estimator is:

$$E\left(\frac{S^2}{n}\right) = BV(\bar{Y}), \quad \text{where } B = \frac{n/c - 1}{n - 1}$$

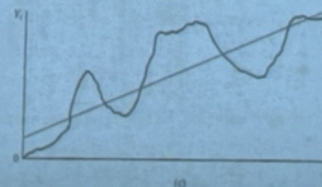
- a) Stationary time series Y_i exhibiting positive autocorrelation.



- b) Stationary time series Y_i exhibiting negative autocorrelation.



- c) Nonstationary time series with an upward trend



11.5.3 Replication Method for Steady-State Simulations

If initialization bias in the point estimator has been reduced to a negligible level (through some combination of intelligent initialization and deletion), then the method of independent replications can be used to estimate point-estimator variability and to construct a confidence interval. The basic idea is simple: Make R replications, initializing and deleting from each one the same way.

If, however, significant bias remains in the point estimator and a large number of replications are used to reduce point-estimator variability, the resulting confidence interval can be misleading. This happens because bias is not affected by the number of replications R ; it is affected only by deleting more data (i.e., increasing T_0) or extending the length of each run (i.e., increasing T_E). Thus, increasing the number of replications R could produce shorter confidence intervals around the "wrong point." Therefore, it is important to do a thorough job of investigating the initial-condition bias.

If the simulation analyst decides to delete d observations of the total of n observations in a replication, then the point estimator of θ is

$$\bar{Y}_{..}(n, d) = \frac{1}{R} \sum_{r=1}^R \frac{1}{n-d} \sum_{j=d+1}^n Y_{rj} \quad (11.29)$$

That is, the point estimator is the average of the remaining data. The basic raw output data $\{Y_{rj}, r = 1, \dots, R; j = 1, \dots, n\}$ are exhibited in Table 11.4. For instance, Y_{rj} could be the delay of customer j in queue, or the response time of job j in a job shop, on replication r . The number d of deleted

Table 11.4 Raw Output Data from a Steady-State Simulation

Replication	Observations						Replication Averages
	1	...	d	$d+1$	...	n	
1	$Y_{1,1}$	...	$Y_{1,d}$	$Y_{1,d+1}$	...	$Y_{1,n}$	$\bar{Y}_{1..}(n, d)$
2	$Y_{2,1}$	...	$Y_{2,d}$	$Y_{2,d+1}$	...	$Y_{2,n}$	$\bar{Y}_{2..}(n, d)$
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
R	$Y_{R,1}$	...	$Y_{R,d}$	$Y_{R,d+1}$	...	$Y_{R,n}$	$\bar{Y}_{R..}(n, d)$
	$\bar{Y}_{..1}$	...	$\bar{Y}_{..d}$	$\bar{Y}_{..d+1}$	...	$\bar{Y}_{..n}$	$\bar{Y}_{..}(n, d)$

observations and the total number of observations n might vary from one replication to the next, in which case replace d by d_r and n by n_r . For simplicity, assume that d and n are constant over replications.

When using the replication method, each replication is regarded as a single sample for the purpose of estimating θ . For replication r , define

$$\bar{Y}_{r..}(n, d) = \frac{1}{n-d} \sum_{j=d+1}^n Y_{rj}$$

as the sample mean of all (nondeleted) observations in replication r . Because all replications use different random-number streams and all are initialized at time 0 by the same set of initial conditions I_0 , the replication averages

$$\bar{Y}_{1.}(n, d), \dots, \bar{Y}_{R.}(n, d)$$

are independent and identically distributed random variables; that is, they constitute a random sample from some underlying population having unknown mean

$$\theta_{n,d} = E[\bar{Y}_{r.}(n, d)] \quad (11.31)$$

The overall point estimator, given in Equation (11.29), is also given by

$$\bar{Y}_{..}(n, d) = \frac{1}{R} \sum_{r=1}^R \bar{Y}_{r.}(n, d) \quad (11.32)$$

as can be seen from Table 11.4 or from using Equation (11.21). Thus, it follows that

$$E[\bar{Y}_{..}(n, d)] = \theta_{n,d}$$

M/G/1 queueing example:

- Suppose $R = 10$, each of length $T_E = 15,000$ minutes, starting at time 0 in the empty and idle state, initialized for $T_0 = 2,000$ minutes before data collection begins.
- Each batch means is the average number of customers in queue for a 1,000-minute interval.
- The 1st two batch means are deleted ($d = 2$).
- The point estimator and standard error are:

$$\bar{Y}_{..}(15,2) = 8.43 \quad \text{and} \quad s.e.(\bar{Y}_{..}(15,2)) = 1.59$$
- The 95% C.I. for long-run mean queue length is:

$$\bar{Y}_{..} - t_{\alpha/2, R-1} S / \sqrt{R} \leq \theta \leq \bar{Y}_{..} + t_{\alpha/2, R-1} S / \sqrt{R}$$

$$8.43 - 2.26(1.59) \leq L_Q \leq 8.43 + 2.26(1.59)$$
- A high degree of confidence that the long-run mean queue length is between **4.84 and 12.02** (if d and n are "large" enough).

Sample Size: A specified precision can be achieved in steady state simulation by:

- Increasing the number of replications
- Increasing the total length

An alternative to increasing R is to increase total run length $T_0 + T_E$ within each replication:

- Increase run length from $(T_0 + T_E)$ to $(R/RO)(T_0 + T_E)$, and
- Delete additional amount of data, from time 0 to time $(R/RO) T_0$ Advantage: any residual bias in the point estimator should be further reduced.

However, it is necessary to have saved the state of the model at time $T_0 + T_E$ and to be able to restart the model.

Batch Means for Interval Estimation

[Steady-State Simulations]

- Using a single, long replication:
 - Problem: data are dependent so the usual estimator is biased.
 - Solution: batch means.
- Batch means: divide the output data from 1 replication (after appropriate deletion) into a few large batches and then treat the means of these batches as if they were independent.
- A continuous-time process, $\{Y(t), T_0 \leq t \leq T_0 + T_E\}$:
 - k batches of size $m = T_E/k$, batch means:

$$\bar{Y}_j = \frac{1}{m} \int_{(j-1)m}^{jm} Y(t + T_0) dt$$

A discrete-time process, $\{Y_i, i = d+1, d+2, \dots, n\}$:

- k batches of size $m = (n - d)/k$, batch means: $\bar{Y}_j = \frac{1}{m} \sum_{i=(j-1)m+1}^{jm} Y_{i+d}$

$$\underbrace{Y_1, \dots, Y_d}_{\text{deleted}}, \underbrace{Y_{d+1}, \dots, Y_{d+m}}_{\bar{Y}_1}, \underbrace{Y_{d+m+1}, \dots, Y_{d+2m}}_{\bar{Y}_2}, \dots, \underbrace{Y_{d+(k-1)m+1}, \dots, Y_{d+km}}_{\bar{Y}_k}$$

- Starting either with continuous-time or discrete-time data, the variance of the sample mean is estimated by:

$$\frac{S^2}{k} = \frac{1}{k} \sum_{j=1}^k \frac{(\bar{Y}_j - \bar{Y})^2}{k-1} = \sum_{j=1}^k \frac{\bar{Y}_j^2 - k\bar{Y}^2}{k(k-1)}$$

- If the batch size is sufficiently large, successive batch means will be approximately independent, and the variance estimator will be approximately unbiased.

No widely accepted and relatively simple method for choosing an acceptable batch size m (see text for a suggested approach). Some simulation software does it automatically.

Batch Means for Interval Estimation

- Number of batches between 10 to 30
- Lag-1 autocorrelation is usually studied to assess the dependence between batch means.
- If it is zero, then batch means are treated as independent.
- Good strategy is to allow the number of batches to increase as the square root of sample size after finding a batch size for which lag-1 autocorrelation is approximately 0.

Atomicity: Atomicity ensures that a transaction (a set of operations) is either fully completed or not done at all.

- When modeling systems that simulate transactions (e.g., banking systems, order processing), atomicity ensures consistency in simulation state.

- If a simulation fails mid-operation (say, transferring money between accounts), atomicity ensures the system rolls back and maintains correctness.
- Helps in discrete-event simulations when multiple events act on shared states.

Log-based Recovery: This technique uses logs (a sequential record of actions) to restore a system to a consistent state after a failure.

- Used in simulations with checkpoints or rollback mechanisms.
- Example: In time-warp simulation protocols (used in parallel discrete event simulation), logs help rollback a simulation state when a causality violation is detected.
- Log-based recovery ensures minimal recomputation by replaying only necessary actions.

Shadow Paging: Instead of writing changes directly to the main data, it writes to a "shadow copy." Once the transaction is successful, it switches the shadow copy as the main version.

- Useful in simulations that deal with file-system behavior, transactional systems, or persistent memory models.
- Shadow paging eliminates the need for undo operations, which can simplify recovery in simulations.
- Can model crash-recovery systems (e.g., embedded systems or databases in flight control simulations).

Monte Carlo Estimation

Monte Carlo methods approximate complex integrals or expected values by simulating random samples. For a continuous function $g(\theta)$ over interval $[a, b]$, the integral is rewritten as an expected value with respect to a uniform distribution $U \sim [a, b]$:

$$\int_a^b g(\theta) d\theta = (b - a) \cdot E[g(U)]$$

This expectation is approximated using the method of moments estimator:

$$E[g(X)] \approx \frac{1}{n} \sum_{i=1}^n g(X_i)$$

Finite-Horizon vs. Infinite-Horizon Statistics

Finite-Horizon (Transient) Statistics

- Simulate systems over a finite time τ .
- Affected by initial conditions (e.g., an initially idle server) ².
- Example: Simulating 100 jobs in an M/M/1 queue with arrival rate $\lambda = 1.0$ and service rate $\nu = 1.25$ yields an average wait of 3.33 ± 1.04 (95% confidence interval) ².

Infinite-Horizon (Steady-State) Statistics

- Aim to estimate long-term behavior as $\tau \rightarrow \infty$.
- Initial conditions become irrelevant (e.g., inventory levels stabilizing at 26.75 after 2560 intervals) ².